WEEK 9 PROJECT REPORT

HOUSING DATA CLEANING & TRANSFORMATION

Name: Santhiya M

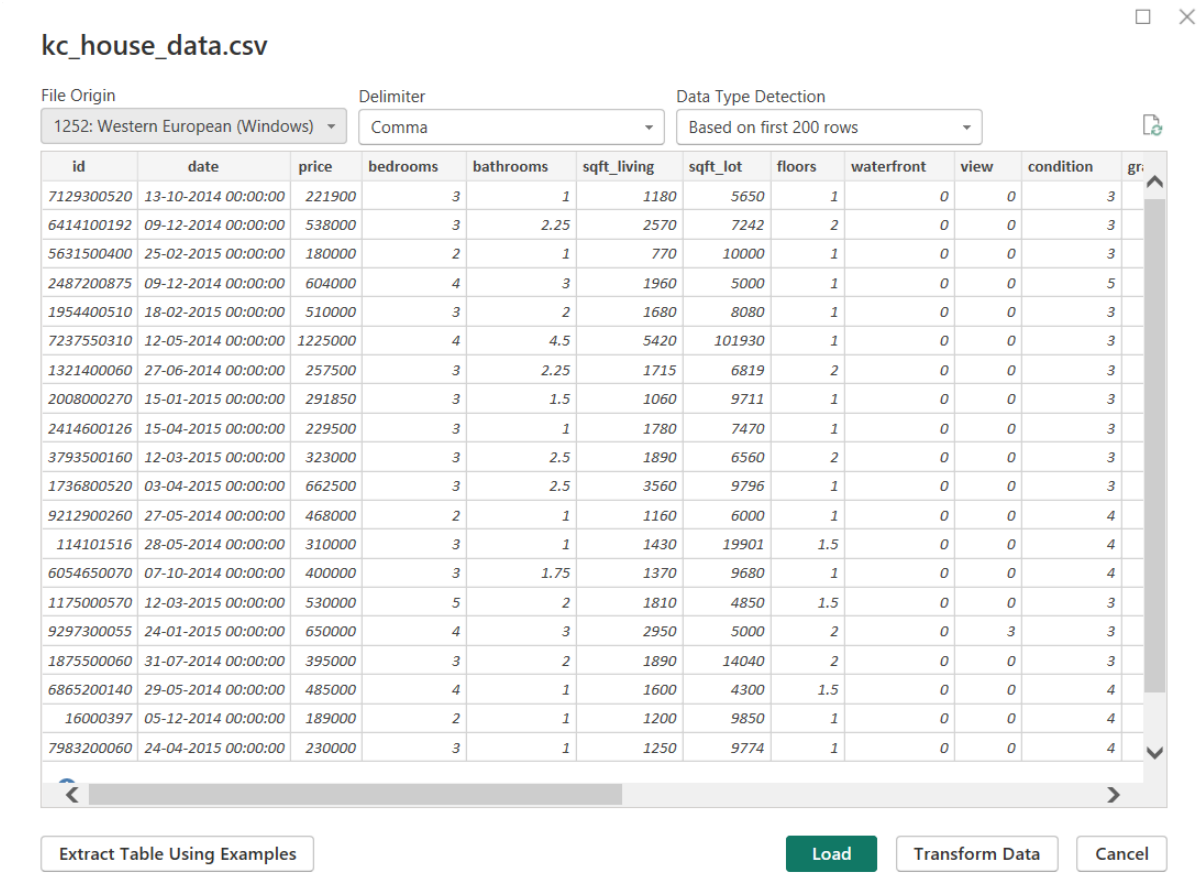
Project Title: Housing Data Cleaning and Transformation using Power BI
Tool Used: Microsoft Power BI Desktop
Dataset: King County House Sales Dataset

INTRODUCTION

The objective of Week 9 was to perform data cleaning and transformation on the King County House Sales Dataset using Microsoft Power BI. The dataset contains information about house prices, property sizes, bedrooms, bathrooms, construction years, renovation years, and sale dates. The purpose of this task was to prepare accurate and structured data for analyzing historical house sales and understanding the factors that influence property prices.

DATASET IMPORT

The King County House Sales Dataset was imported into Power BI Desktop using the Get Data option. The data was opened in Power Query Editor, where the columns, data types, and overall data structure were examined before beginning the cleaning process.



kc_house_data.csv

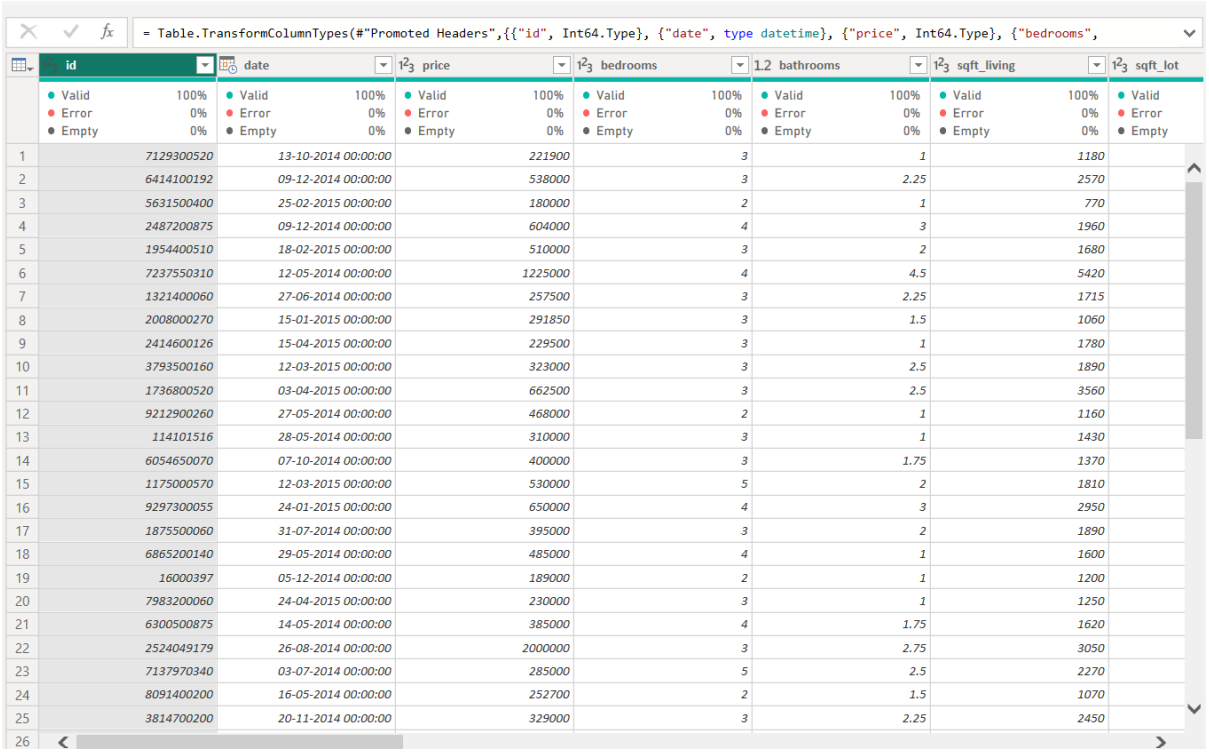
File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	gr
7129300520	13-10-2014 00:00:00	221900	3	1	1180	5650	1	0	0	3	
6414100192	09-12-2014 00:00:00	538000	3	2.25	2570	7242	2	0	0	3	
5631500400	25-02-2015 00:00:00	180000	2	1	770	10000	1	0	0	3	
2487200875	09-12-2014 00:00:00	604000	4	3	1960	5000	1	0	0	5	
1954400510	18-02-2015 00:00:00	510000	3	2	1680	8080	1	0	0	3	
7237550310	12-05-2014 00:00:00	1225000	4	4.5	5420	101930	1	0	0	3	
1321400060	27-06-2014 00:00:00	257500	3	2.25	1715	6819	2	0	0	3	
2008000270	15-01-2015 00:00:00	291850	3	1.5	1060	9711	1	0	0	3	
2414600126	15-04-2015 00:00:00	229500	3	1	1780	7470	1	0	0	3	
3793500160	12-03-2015 00:00:00	323000	3	2.5	1890	6560	2	0	0	3	
1736800520	03-04-2015 00:00:00	662500	3	2.5	3560	9796	1	0	0	3	
9212900260	27-05-2014 00:00:00	468000	2	1	1160	6000	1	0	0	4	
114101516	28-05-2014 00:00:00	310000	3	1	1430	19901	1.5	0	0	4	
6054650070	07-10-2014 00:00:00	400000	3	1.75	1370	9680	1	0	0	4	
1175000570	12-03-2015 00:00:00	530000	5	2	1810	4850	1.5	0	0	3	
9297300055	24-01-2015 00:00:00	650000	4	3	2950	5000	2	0	3	3	
1875500060	31-07-2014 00:00:00	395000	3	2	1890	14040	2	0	0	3	
6865200140	29-05-2014 00:00:00	485000	4	1	1600	4300	1.5	0	0	4	
16000397	05-12-2014 00:00:00	189000	2	1	1200	9850	1	0	0	4	
7983200060	24-04-2015 00:00:00	230000	3	1	1250	9774	1	0	0	4	

Extract Table Using Examples | Load | Transform Data | Cancel

DATA CLEANING

The dataset was inspected for duplicate records, missing values, incorrect data types, and invalid entries. Duplicate house records were checked using the unique house identifier, and duplicate identifiers were removed where appropriate. Missing and invalid values in important columns, including price, bedrooms, bathrooms, and living area, were reviewed and handled. Records with zero or invalid bedroom and bathroom counts were filtered out to improve data quality.

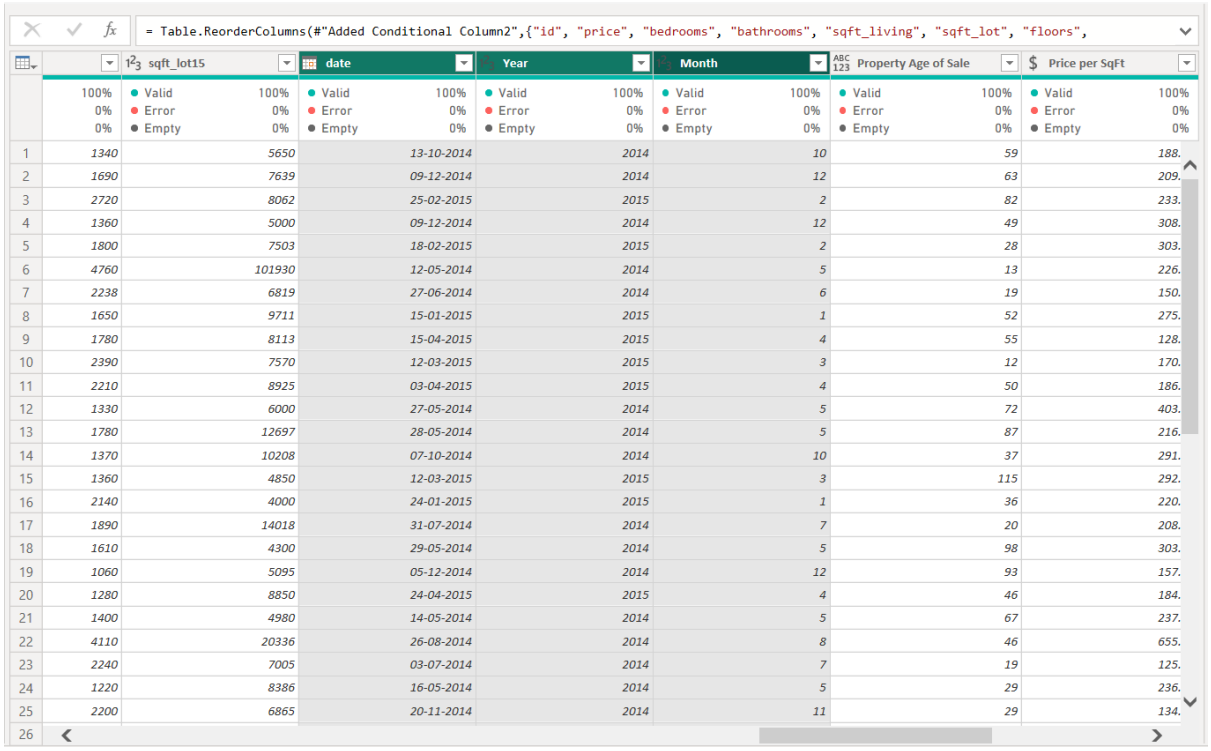


= Table.TransformColumnTypes(#"Promoted Headers",{{"id", Int64.Type}, {"date", type datetime}, {"price", Int64.Type}, {"bedrooms",

	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	7129300520	13-10-2014 00:00:00	221900	3	1	1180	
2	6414100192	09-12-2014 00:00:00	538000	3	2.25	2570	
3	5631500400	25-02-2015 00:00:00	180000	2	1	770	
4	2487200875	09-12-2014 00:00:00	604000	4	3	1960	
5	1954400510	18-02-2015 00:00:00	510000	3	2	1680	
6	7237550310	12-05-2014 00:00:00	1225000	4	4.5	5420	
7	1321400060	27-06-2014 00:00:00	257500	3	2.25	1715	
8	2008000270	15-01-2015 00:00:00	291850	3	1.5	1060	
9	2414600126	15-04-2015 00:00:00	229500	3	1	1780	
10	3793500160	12-03-2015 00:00:00	323000	3	2.5	1890	
11	1736800520	03-04-2015 00:00:00	662500	3	2.5	3560	
12	9212900260	27-05-2014 00:00:00	468000	2	1	1160	
13	114101516	28-05-2014 00:00:00	310000	3	1	1430	
14	6054650070	07-10-2014 00:00:00	400000	3	1.75	1370	
15	1175000570	12-03-2015 00:00:00	530000	5	2	1810	
16	9297300055	24-01-2015 00:00:00	650000	4	3	2950	
17	1875500060	31-07-2014 00:00:00	395000	3	2	1890	
18	6865200140	29-05-2014 00:00:00	485000	4	1	1600	
19	16000397	05-12-2014 00:00:00	189000	2	1	1200	
20	7983200060	24-04-2015 00:00:00	230000	3	1	1250	
21	6300500875	14-05-2014 00:00:00	385000	4	1.75	1620	
22	2524049179	26-08-2014 00:00:00	2000000	3	2.75	3050	
23	7137970340	03-07-2014 00:00:00	285000	5	2.5	2270	
24	8091400200	16-05-2014 00:00:00	252700	2	1.5	1070	
25	3814700200	20-11-2014 00:00:00	329000	3	2.25	2450	
26							

DATE TRANSFORMATION

The original sale date column was converted into a usable Date format. Two additional columns, Sale Year and Sale Month, were created to extract temporal information from the sale dates. These columns make it easier to organize and analyze house sales by year and month.



```
= Table.ReorderColumns(#"Added Conditional Column2",{"id", "price", "bedrooms", "bathrooms", "sqft_living", "sqft_lot", "floors",
```

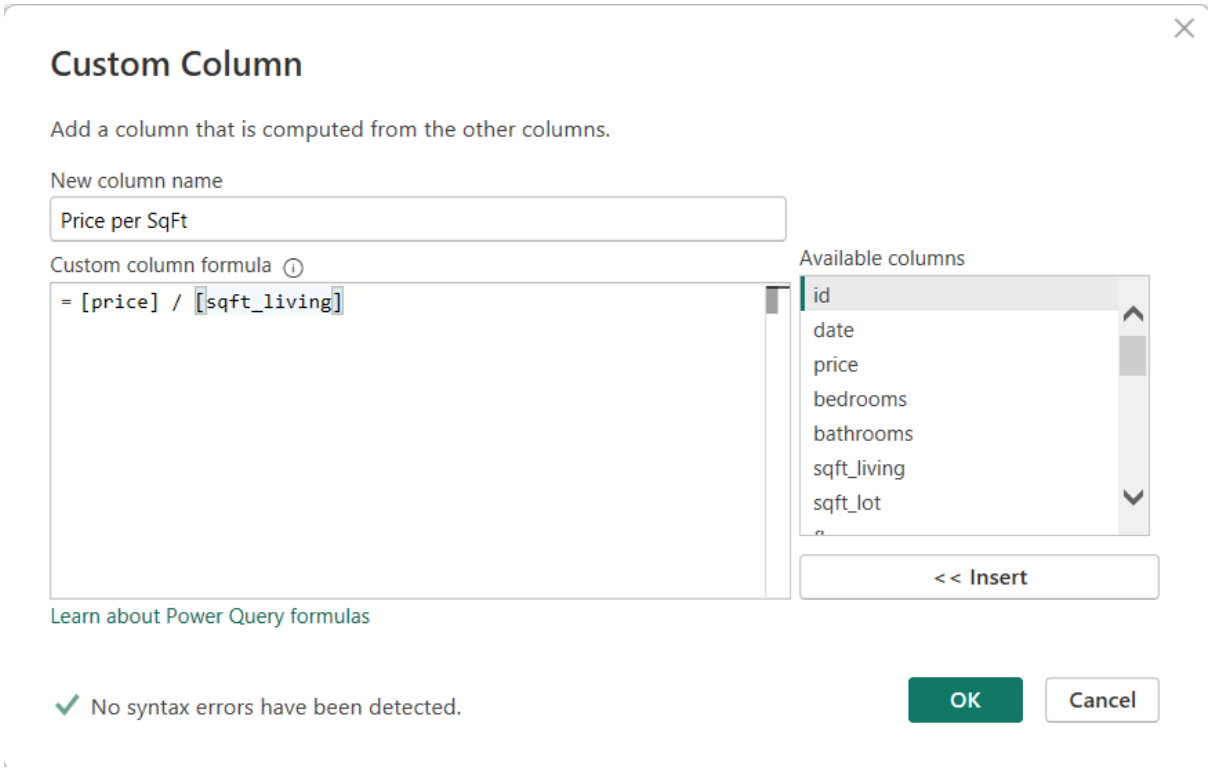
	sqft_lot15	date	Year	Month	Property Age of Sale	Price per SqFt
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
1	5650	13-10-2014	2014	10	59	188.
2	7639	09-12-2014	2014	12	63	209.
3	8062	25-02-2015	2015	2	82	233.
4	5000	09-12-2014	2014	12	49	308.
5	7503	18-02-2015	2015	2	28	303.
6	101930	12-05-2014	2014	5	13	226.
7	6819	27-06-2014	2014	6	19	150.
8	9711	15-01-2015	2015	1	52	275.
9	8113	15-04-2015	2015	4	55	128.
10	7570	12-03-2015	2015	3	12	170.
11	8925	03-04-2015	2015	4	50	186.
12	6000	27-05-2014	2014	5	72	403.
13	12697	28-05-2014	2014	5	87	216.
14	10208	07-10-2014	2014	10	37	291.
15	4850	12-03-2015	2015	3	115	292.
16	4000	24-01-2015	2015	1	36	220.
17	14018	31-07-2014	2014	7	20	208.
18	4300	29-05-2014	2014	5	98	303.
19	5095	05-12-2014	2014	12	93	157.
20	8850	24-04-2015	2015	4	46	184.
21	4980	14-05-2014	2014	5	67	237.
22	20336	26-08-2014	2014	8	46	655.
23	7005	03-07-2014	2014	7	19	125.
24	8386	16-05-2014	2014	5	29	236.
25	6865	20-11-2014	2014	11	29	134.
26						

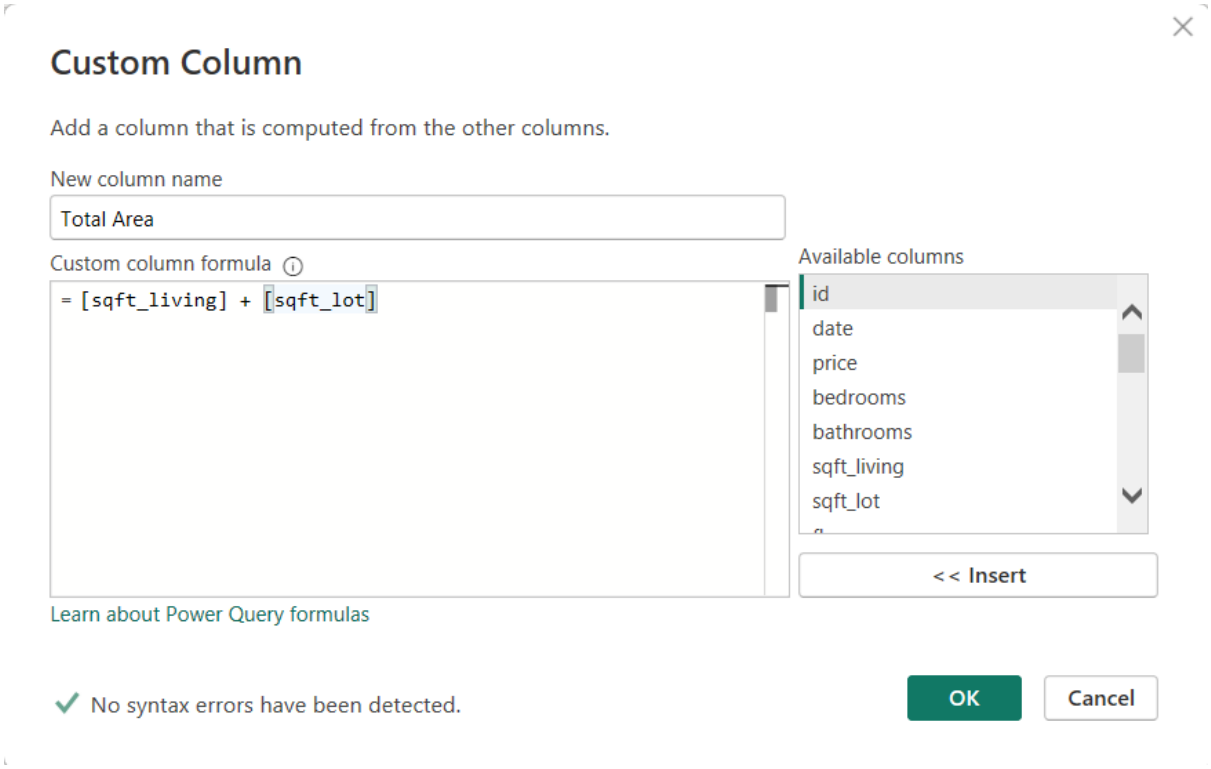
CREATION OF CALCULATED COLUMNS

Several calculated columns were added in Power Query to derive useful information from the existing dataset.

- **Property Age of Sale:** Calculated by subtracting the construction year from the year of sale.
- **Price per SqFt:** Calculated by dividing the house sale price by its living area in square feet.
- **Total Area:** Created by combining the living area and lot area, following the selected definition for this task.

These calculations provide additional information for comparing properties and examining the relationship between property characteristics and sale prices.


Custom Column
Add a column that is computed from the other columns.
New column name
Price per SqFt
Custom column formula
= [price] / [sqft_living]
Available columns
id
date
price
bedrooms
bathrooms
sqft_living
sqft_lot
<< Insert
Learn about Power Query formulas
No syntax errors have been detected.
OK
Cancel

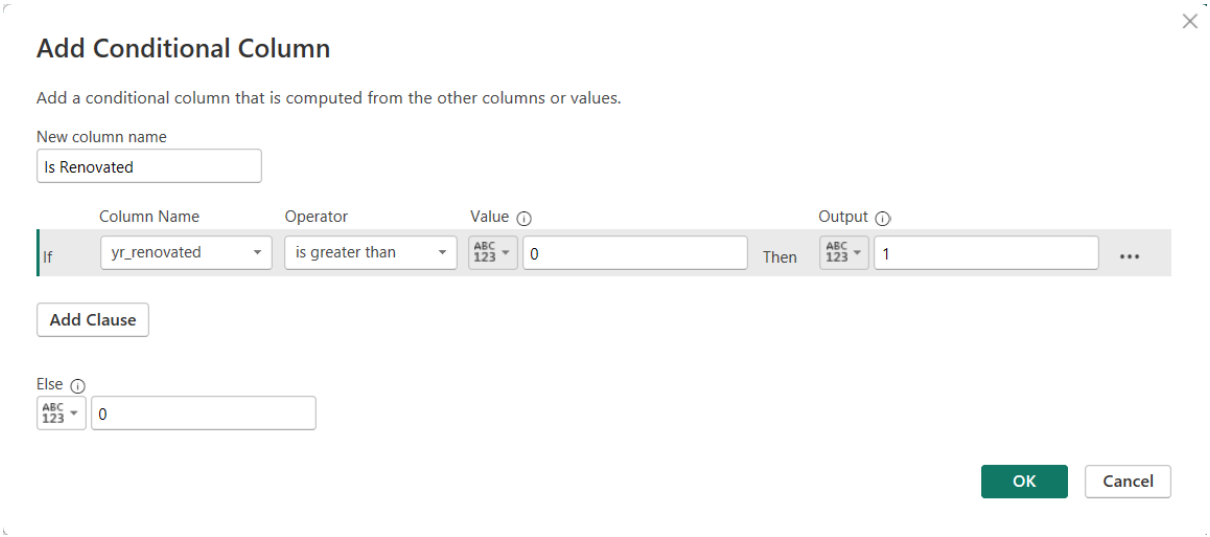

Custom Column
Add a column that is computed from the other columns.
New column name
Total Area
Custom column formula
= [sqft_living] + [sqft_lot]
Available columns
id
date
price
bedrooms
bathrooms
sqft_living
sqft_lot
<< Insert
Learn about Power Query formulas
No syntax errors have been detected.
OK
Cancel

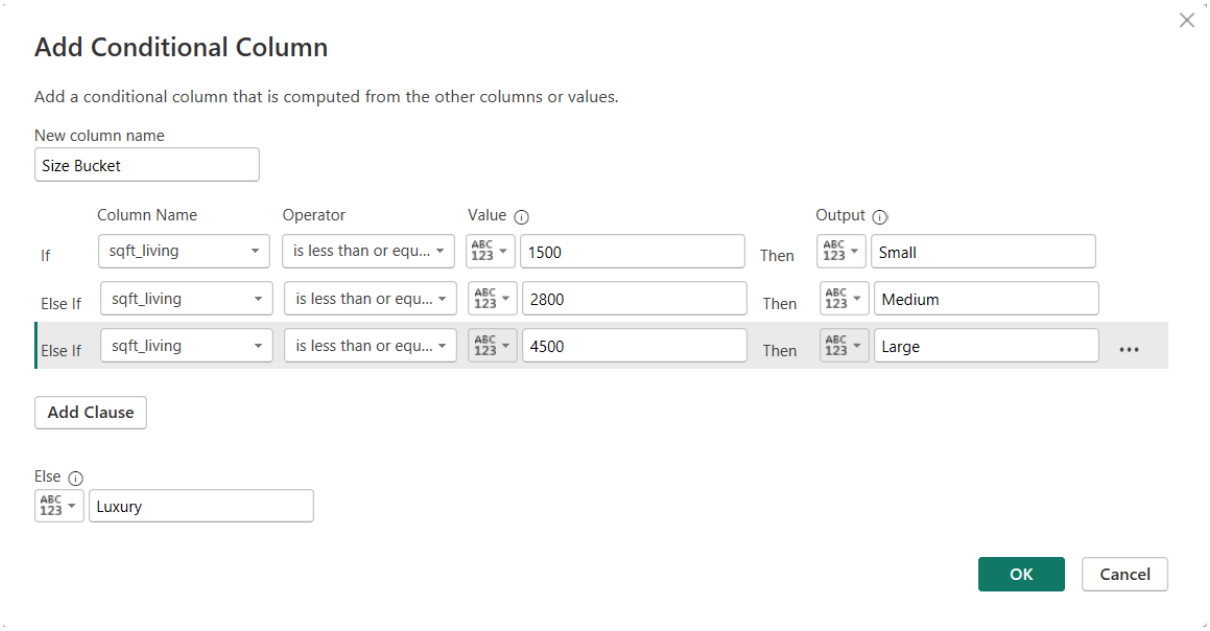
PROPERTY CATEGORIZATION

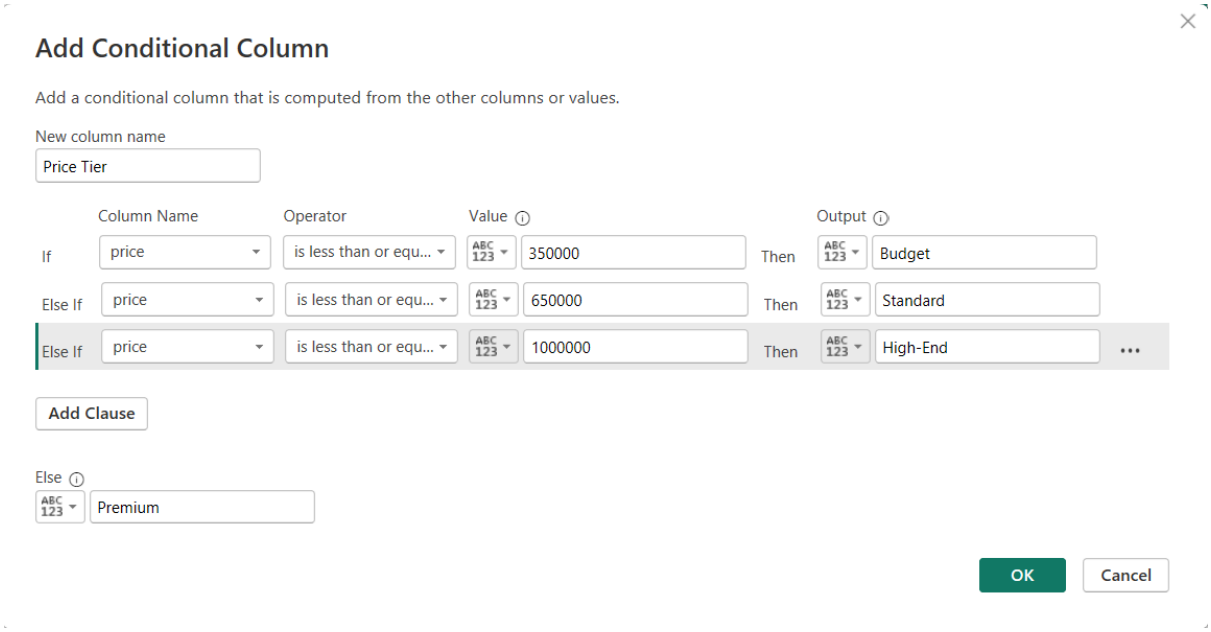
Additional columns were created to categorize properties and simplify analysis.

- **Is Renovated:** Identifies whether a property has a recorded renovation year.
- **Size Bucket:** Categorizes properties into Small, Medium, Large, and Luxury groups based on living area.
- **Price Tier:** Classifies houses into Budget, Mid-Range, Premium, and Luxury categories according to their sale prices.

These categories make the dataset easier to interpret and support comparisons between different types of properties.







DATA VALIDATION AND FINAL LOADING

After completing the transformations, the dataset was reviewed to check the accuracy of the new columns, data types, and filtered records. The transformed data was then loaded into Power BI using the Close & Apply option. The final table was checked to ensure that the cleaned records and newly created columns were available for further analysis.

CONCLUSION

The Week 9 task provided practical experience in housing data cleaning and transformation using Microsoft Power BI. The dataset was inspected, cleaned, and enhanced with date attributes, calculated columns, and property categories. These transformations improved the structure and usability of the data, making it more suitable for further analysis of housing prices and property characteristics.